scientific reports

Check for updates

OPEN

Optimization design and torque performance research of interior permanent magnet synchronous motors

Lianbo Niu

The high magnetic saturation at flux isolation bridge of interior permanent magnet synchronous motor (IPMSM) enriches the harmonic content of air gap flux density. When IPMSM operates at low speeds, these harmonics of air gap flux density increase torque ripple. An IPMSM with I2V type is proposed to further improve sine degree of no-load air gap flux density, achieving high torque density and low torque ripple. The V type, IV type and I2V type are selected as the comparative objects for optimization research, the established 2D parameterized finite element analysis (FEA) models are used to optimize and analyze opening angle and pole arc angle of the first layer of permanent magnet, which have the greatest impact on the electromagnetic performance of IPMSMs. Under the constraint of same volume of permanent magnets, the optimization models of three different rotor structures are determined ultimately. The no-load air gap flux density waveforms and harmonic contents of IPMSM with three different optimized rotor topology structures are analyzed and compared, the results show that the no-load air gap flux density of I2V type proposed in this paper have the lowest total harmonic distortion rate (THD).The torque performances are analyzed and compared with the other two, the results show that torque performances of I2V type are superior to the other two as the amount of permanent magnet is slightly reduced. Aiming at problems that the torque ripple of I2V type is greater than the other two under rated operating conditions, this paper proposes a simple and effective method to add a set of circular flux isolation holes under each pole of the rotor, and the position and diameter of flux isolation holes are adjusted to further optimize the magnetic circuit in order to reduce torque ripple. The optimization results showed that reduction ratio of torque ripple with optimized I2V type is 67.3% under rated operating conditions. The I2V prototype is processed and manufactured and a prototype experimental platform is built. Experimental verifications are conducted on the torque, external characteristics. The test results are basically consistent with the FEA results, proving the accuracy of the analysis and modeling, as well as the effectiveness of simulation calculations in this paper.

Keywords Interior permanent magnet synchronous motor (IPMSM), Torque ripple, Flux separation hole, Output average torque

At present, electric vehicle drive motors mainly include induction motors, switched reluctance motors, and permanent magnet synchronous motors (PMSMs). The manufacturing process of induction motors is mature, highly reliable, easy to maintain, and has strong environmental adaptability, but the efficiency and the torque density are low under urban working conditions[1]. Switched Reluctance Motors are also promising electric vehicle drive motors, with the advantages of relatively simple structure, high reliability, and low cost as they do not require rare earth permanent magnets. However, their disadvantages include high torque ripple, high noise and relatively poor vehicle comfort[2,3].

PMSMs have the advantages of high torque and power density, high efficiency, small size, and constant power with wide speed regulation, making them particularly suitable for use as drive motors in electric vehicles[4,5]. Due to the difference in d-q inductance, Interior permanent magnet synchronous motor (IPMSM) can fully utilize the reluctance torque by controlling the current angle. The utilization of reluctance torque greatly increases the overload capacity and torque density of the motor, and through flux weakening acceleration, the constant

School of Mechanical and Electronically Engineering, Xinxiang University, Xinxiang 453003, China. email: niulianbo@163.com

power operating range is greatly improved. Therefore, the research on the performance of IPMSMs has become a research hotspot for electric vehicle drive motors[6,7].

Increasing the number of permanent magnet layers in the rotor can reduce the harmonic content of air gap flux density waveform and decrease iron loss; However, the effects of increasing torque density and reducing eddy current loss of permanent magnets are not ideal. The waveform of no-load back electromotive force has large harmonics and poor sinuosity[8]. Xiao Chen et al. analyzed and studied the IPMSMs with a 2 V type rotor topology structure. The research results indicate that due to the limitation of maximum output voltage of inverter, the current waveform distortion and torque ripple significantly increase under deep flux weakening conditions[9]. Wenying Hu conducted a comparative study on IPMSMs with five different rotor topologies, namely V type, U type, 2 V type, IU type, and 2I type. The results showed that double-layer rotor structure had better performance than single-layer rotor structure in terms of permanent magnet utilization, flux-weakening capability, operating efficiency, and anti-demagnetization ability[10]. Yaohua Hu proposed a three-layer rotor topology structure with IVU type and compared its electromagnetic performance with V type and IV type rotor structure. The results show that three-layer rotor topology of IVU type has better performance than the other two in terms of air gap flux density harmonic content, flux-weakening capability and iron loss during high-speed operation. However, the IVU type has a lower torque density than IV type[11].

The above literature indicates that increasing the number of permanent magnet layers and optimizing their rotor topologies can effectively improve the utilization rate of reluctance torque, enhance flux-weakening expansion capability and reduce iron loss. However, it will to reduce the torque density, increase torque ripple, and harmonic content of no-load air gap flux density, making it difficult to simultaneously consider all of the above electromagnetic properties. This paper proposes an IPMSM with I2V type rotor topology structure, and establishes 2D parameterized FEA model to optimize and analyze the rotor structure, achieving high torque density while significantly reducing torque ripple.

Electromagnetic torque analysis

Torque ripple characteristics analysis of PMSM

For IPMSMs, the air gap magnetic resistance is uneven. Under the current excitation of i_s, the generated armature reaction magnetic field will change with the torque angle θ, making it difficult to calculate and analyze the armature reaction magnetic field. In motor analysis, the double reaction theory is often used to analyze the problem of salient pole synchronous motors. For IPMSMs, this analysis method can also be used, and a synchronous rotating d-q coordinate system as shown in Fig. 1 is used to construct the mathematical model of IPMSM.

The air gap magnetic field energy W_m of IPMSM shown in Fig. 1 can be expressed as

$$\begin{aligned} W_m &= \int_V \frac{B^2}{2\mu_0} dv \\ &= \frac{L_{ef} L_g}{4\mu_0} \frac{D}{p} \int_0^{p2\pi} \begin{bmatrix} B_r^2 \cos^2(\omega_e t - \theta_e) + B_s^2 \cos^2(\omega_e t - \theta_e - \delta) \\ +2B_r B_s \cos(\omega_e t - \theta_e) \cos(\omega_e t - \theta_e - \delta) \end{bmatrix} d\theta_e \\ &= \frac{L_{ef} L_g \pi D}{4\mu_0} \left[B_r^2 + B_s^2 + 2B_r B_s \cos(\delta) \right] \end{aligned} \tag{1}$$

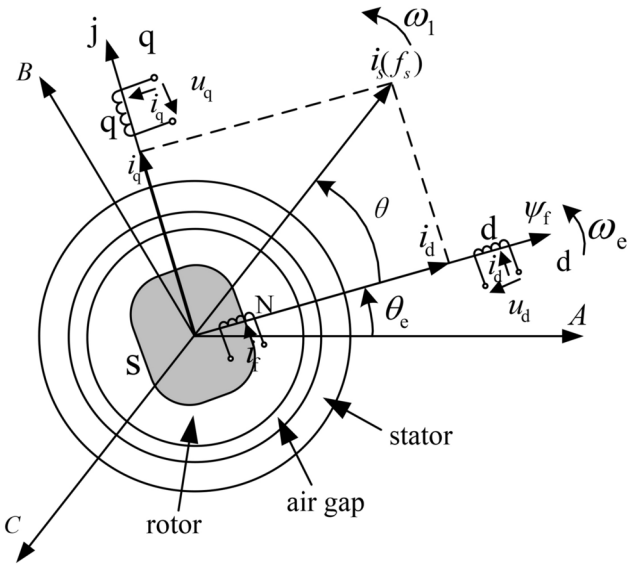


Fig. 1. Physical model of the IPMSM in d-q coordinates.

where, B_s is amplitude of air gap flux density generated by the three-phase stator current, B_r is amplitude of air gap flux density generated by the permanent magnet, and δ is electrical angle at which the fundamental component of B_s leads B_r after Fourier decomposition. L_g is the length of the air gap, L_{ef} is the axial length of the motor, and D is the average diameter of the air gap of the motor. ω_e is rotational electrical angular velocity in d-q coordinates, p is pole pairs, μ_0 is vacuum magnetic permeability, and θ_e is electrical angle at which the d-axis rotates relative to the A-phase space reference axis.

Based on the principle of virtual displacement, the energy of the air gap magnetic field W_m is negatively biased towards the relative position angle δ of the magnetic flux density between the stator and rotor, and the electromagnetic torque can be expressed as

$$T_e = p\frac{\partial W_m}{\partial \delta} = -p\frac{L_{ef}L_g\pi D}{2\mu_0}B_rB_s\sin(\delta) \tag{2}$$

Fourier transform the stator flux density B_s and rotor flux density B_r separately, the torque ripple of the motor can be expressed as

$$T_{rip} = -p\frac{L_{ef}L_g\pi D}{2\mu_0}\sum_{k=1,2,3\ldots} kB_{r,k}B_{s,k}\sin(6h\theta_e \pm \delta) \tag{3}$$

Equation 1 needs further clarification as follows

$$\int_0^{p2\pi}[\cos(\omega_e t - v_1\theta_e)\cos(\omega_e t - v_2\theta_e - \delta)]d\theta_e = 0 \quad v_1 \neq v_2 \tag{4}$$

where, v_1 and v_2 are the harmonic orders of B_s and B_r, respectively. According to Eq. (4), when the harmonic orders of B_s and B_r are not equal, the integral value of this phase is 0, the energy of air gap magnetic field W_m remains constant, and the partial derivative value is 0, that is, the electromagnetic torque is 0. So only when the harmonic order of stator air gap flux density and rotor air gap flux density are equal, will torque ripple occur. According to the symmetry characteristics of waveform of rotor air gap flux density, the harmonics after Fourier decomposition only include odd numbered harmonics such as 1, 3, 5, 7, 9…, while the stator air gap flux density harmonics of IPMSM do not include 3rd order and 3rd harmonic harmonics. Therefore, the torque ripple will only occur when both the harmonic order of rotor air gap flux density and stator air gap flux density are 6 $k \pm 1$.

By Fourier expanding the stator air gap flux density B_s and the rotor air gap flux density B_r, the torque ripple ratio can be expressed as

$$\frac{T_{rip}}{T_{\mathrm{av}}} = \frac{1}{\sin(\theta)}\sum_{k=1,2,3\ldots}(6k \mp 1)\frac{B_{s,6k\mp1}}{B_{s,1}}\frac{B_{r,6k\mp1}}{B_{r,1}}\sin(6k\omega t \pm \theta) \tag{5}$$

where, T_{rip} represents torque ripple, T_{av} represents average torque. θ represents torque angle, which is the angle between the stator space current vector i_s and the d-axis.

Equations (3) and (5) indicate that only when the harmonic order of the flux density in air gap between stator and rotor is equal and both are 6 k ± 1, such as the 5th, 7th, 11th, and 13th harmonic orders, will the PMSM generate torque ripple. The magnitude of torque ripple is proportional to harmonic order of flux density, and the higher the harmonic order, the greater the torque ripple. Therefore, increasing the sinusoidal waveform of air gap flux density can effectively reduce torque ripple. After stator structure and stator winding are determined, the stator air gap flux density is basically determined. This article will optimize the key parameters of rotor topology structure to reduce the harmonic distortion rate of air gap flux density, improve its waveform sinuosity, and optimize and reduce the torque ripple.

Steady state analysis of electromagnetic torque of IPMSM in d-q coordinate system

The vector diagram in the d-q coordinate system during steady-state operation of IPMSM is shown in Fig. 2. The angle between stator voltage space vector and q-axis is the voltage angle β. The angle between stator current space vector and q-axis is the current angle α, which ranges from -90° to 90°. The power factor angle φ is angle between stator voltage space vector and stator current space vector. The composite magnetic flux linkage generated by permanent magnet flux linkage ψ_{m} and magnetic flux linkage ψ_{a} generated by stator current is ψ_s, which lags behind stator voltage space vector u_s by an electrical angle of 90°.

According to Fig. 2, the transient voltage of IPMSM in d-q coordinate system can be expressed as

$$\begin{cases} u_d = R_s i_d + L_d\dfrac{\mathrm{d}}{\mathrm{dt}}i_d - \omega_e L_q i_q \\ u_q = R_s \mathrm{i}_q + L_q\dfrac{\mathrm{d}}{\mathrm{dt}}i_q + \omega_e L_d i_d + \omega_e\psi_{\mathrm{m}} \end{cases} \tag{6}$$

The steady-state voltage equation can be expressed as

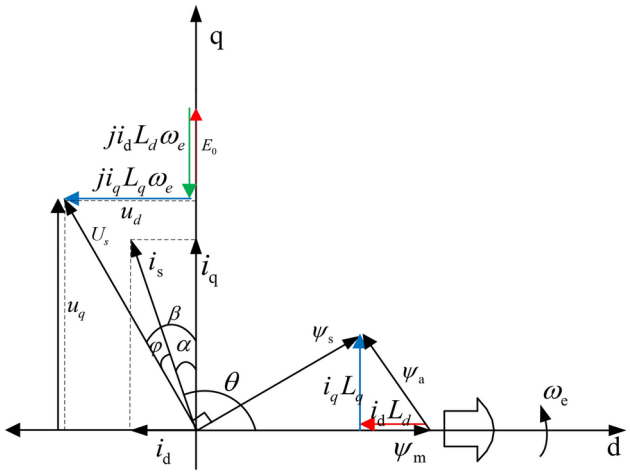


Fig. 2. Phasor diagram of IPMSM in d-q coordinate system.

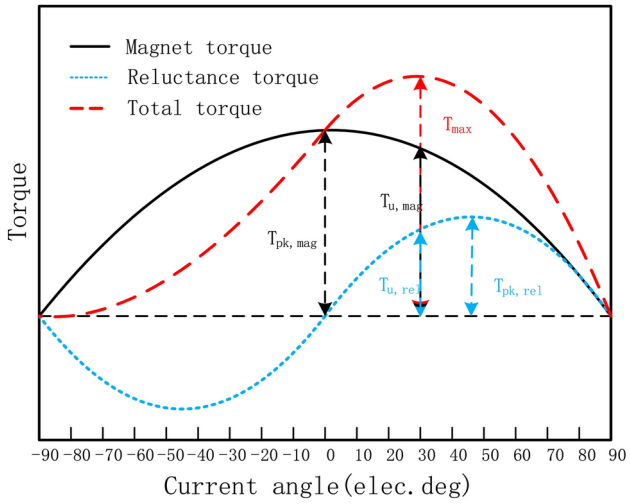


Fig. 3. Torque versus current angle characteristics.

$$\begin{cases} u_d = R_s i_d - \omega_e L_q i_q \\ u_q = R_s i_q + \omega_e L_d i_d + \omega_e \psi_m \end{cases} \tag{7}$$

Due to the small internal resistance, ignoring the voltage drop caused by internal resistance, steady-state voltage can be further expressed as

$$\begin{cases} u_d = -\omega_e L_q i_q \\ u_q = \omega_e L_d i_d + \omega_e \psi_m \end{cases} \tag{8}$$

The power equation of the d-q coordinate system under equal amplitude transformation is shown in Eq. (9), and d-axis current i_d and q-axis current i_q are shown in Eq. (10). The output torque equation can be obtained from Eqs. (9), (10), (11) as shown in Eq. (12). The relationship between the synthesized torque T_u, permanent magnet torque T_{mag}, reluctance torque T_{rel}, and current angle α obtained from Eq. (12). The range of current angle values is -90° to 90°. Permanent magnet torque T_{mag} is proportional to the cosine of the current angle, and Reluctance torque T_{rel} is proportional to the sine of twice the current angle. Generally, when the current angle value is between 30° and 35°, the synthesized torque is the biggest. The relationship between permanent magnet torque, reluctance torque, composite torque and current angle is shown in Fig. 3. L_d and L_q are the equivalent inductance of IPMSM on the d and q axes, and ψ_m is the synthetic magnetic flux generated by the permanent magnet in the d-axis direction.

$$P = \frac{3}{2}(u_d i_d + u_q i_q) \tag{9}$$

$$\begin{cases} i_d = -i_s \sin\alpha \\ i_q = i_s \cos\alpha \end{cases} \tag{10}$$

$$\omega_e = p\Omega \tag{11}$$

$$T_u = 1.5p * \psi_{\mathrm{m}} i_s \cos\alpha + 0.75p * (L_q - L_d) i_s^2 \sin 2\alpha \tag{12}$$

$$T_{mag} = 1.5p * \psi_{\mathrm{m}} i_s \cos\alpha \tag{13}$$

$$T_{rel} = 0.75p * (L_q - L_d) i_s^2 \sin 2\alpha \tag{14}$$

$$T_u = T_{mag} + T_{rel} \tag{15}$$

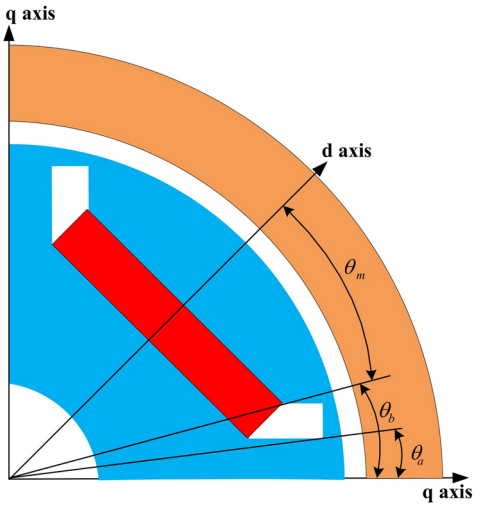
Analysis of air gap flux density

As mentioned above, increasing the sine degree of the waveform and reducing the harmonic distortion rate of the air gap flux density can not only effectively reduce torque ripple, but also effectively reduce iron loss and improve the operating efficiency of IPMSM. Therefore, it is necessary to analyze the air gap flux density and optimize the rotor structure that affects the waveform distribution of air gap flux density.

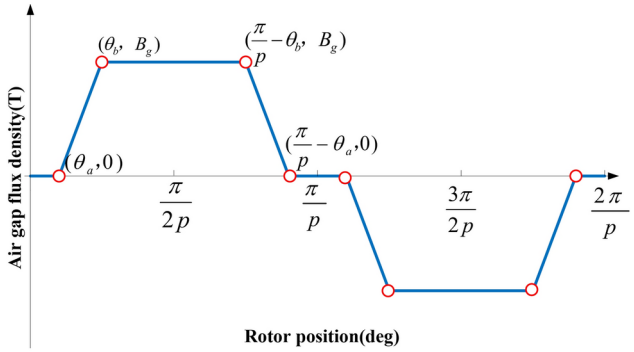
Analysis of no-load air gap flux density of single-layer rotor topologies

Typical single-layer rotor topologies of IPMSMs include I type and V type.

Assuming that the magnetic permeability of the stator and rotor iron cores is infinite, ignoring the motor leakage and magnetic saturation cogging effects. The flux density distribution of no-load air gap with single-layer I Type is shown in Fig. 4. In Fig. 4a, θ_a is the mechanical angle between the q-axis and the starting point of the magnetic barrier, θ_b is the mechanical angle between the q-axis and the end point of the magnetic barrier, and θ_m is the mechanical angle between the d-axis of the permanent magnet and the end point of the magnetic barrier. The flux density of the permanent magnet in the air gap is constant in the region of θ_m, and the air gap flux density is approximately zero in the region where θ_a is close to the q-axis. The air gap flux density increases



(a)Schematic diagram of I type



(b) Waveform of no-load air gap flux density

Fig. 4. Air gap flux density distribution of I type.

continuously from the starting point to the end point of the θ_b magnetic barrier, so it can be regarded as a linear function in the θ_b region of the magnetic barrier. The no-load air gap flux density under the excitation of permanent magnet alone is shown in Eq. (16), and the corresponding waveform is shown in Fig. 4b.

$$B_g(\theta) = \begin{cases} 00 \le \theta < \theta_a \\ \dfrac{B_g}{\theta_b - \theta_a}(\theta - \theta_a)\ \theta_a \le \theta < \theta_b \\ B_g\ \theta_b \le \theta < \dfrac{\pi}{p} - \theta_b \\ -\dfrac{B_g}{\theta_b - \theta_a}\left(\theta - \left(\dfrac{\pi}{p} - \theta_a\right)\right)\ \dfrac{\pi}{p} - \theta_b \le \theta < \dfrac{\pi}{p} - \theta_a \\ 0\ \dfrac{\pi}{p} - \theta_a \le \theta < \dfrac{\pi}{p} \end{cases} \tag{16}$$

The Fourier transform of Eq. (16) can be expressed as

$$B_{gn} = \frac{4B_g}{\pi p(\theta_b - \theta_a)(2n-1)^2}(\sin p\theta_b(2n-1) - \sin p\theta_a(2n-1)) \tag{17}$$

$$B_{gn}(\theta, \alpha_r) = \sum_{n=1}^{\infty} B_{gn} \sin p(\theta - a_r)(2n-1) \tag{18}$$

where, B_g, n, α_r, B_{gn} and p respectively represent the average no-load air gap flux density, harmonic order, rotor angular position, Fourier transform amplitude of air gap flux density, and number of rotor poles.

$$K_l = \frac{2R_m \| R_{bl}}{2R_m} \tag{19}$$

$$\phi_g = \frac{\frac{1}{2K_r R_g}}{\frac{1}{2K_r R_g} + \frac{1}{R_b} + \frac{1}{2K_l R_m}} \phi_{r1} \tag{20}$$

$$B_g = \frac{\phi_g}{A_g} \tag{21}$$

The equivalent magnetic circuit of I type rotor topology is shown in Fig. 5. Compared with the magnetic resistance of air gap, magnetic barrier and permanent magnet, the magnetic resistance of stator and rotor iron cores is sufficiently small, so the magnetic resistance of stator and rotor iron cores is ignored in following analysis. The leakage coefficient of the magnetic barrier is shown in Eq. (19). Applying the law of magnetic flux distribution, the air gap flux is shown in Eq. (20), and the corresponding air gap flux density is shown in Eq. (21). Here, R_m, R_l and R_b represent the magnetic resistance of permanent magnet, magnetic barrier and magnetic bridge, respectively. Φ_g and Φ_{rl} are air gap flux and residual flux of the permanent magnet, respectively.

The equivalent magnetic circuit of V type rotor topology structure is shown in Fig. 6. Compared with I type rotor topology structure, the two q-axis magnetic circuits of V type structure are more uniform, which can effectively improve q-axis inductance, increase salient pole ratio, and improve utilization rate of reluctance torque. In addition, compared with I type topology structure, the size of the opening angle of V type permanent magnet determines the spatial distribution of no-load air gap flux density, which in turn affects its harmonic distortion rate. By adjusting the opening angle of permanent magnet, more permanent magnets can be placed

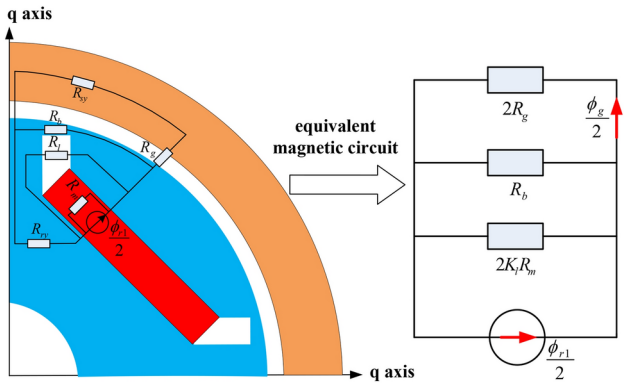


Fig. 5. Magnetic circuit of I type.

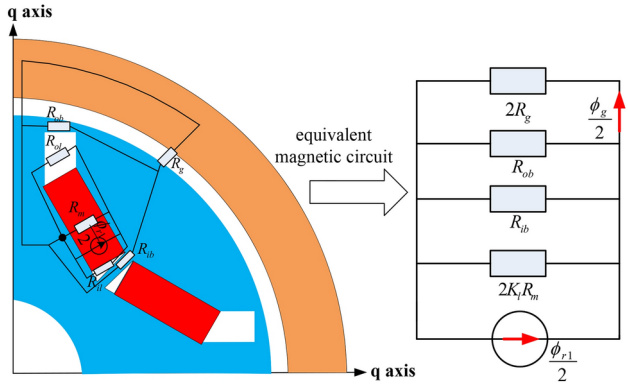


Fig. 6. Magnetic circuit of V type.

first, which is beneficial for magnetic focusing and increases output torque of the permanent magnet, greatly improving the peak power; Secondly, it can improve the sinusoidal waveform of no-load air gap flux density.

The V type topology structure adds internal magnetic barriers and internal magnetic bridges (reinforcing ribs) on both sides of the rotor center axis (d-axis). The V type topology structure adds internal magnetic barriers and magnetic bridges (reinforcing ribs) on both sides of the rotor center axis (d-axis), improving the mechanical strength of rotor structure. When the area of air gap and permanent magnet are the same, compared with I type, the leakage flux generated by the presence of internal magnetic bridges and magnetic barriers in the V type, as shown in Eqs. (22) and (23), leads to a decrease in magnetic flux per pole, which is not conducive to improving electromagnetic output torque, but is beneficial for flux weakening speed regulation.

$$K_{\mathrm{ltotal}} = \frac{2R_m \,\|R_{il}\,\|R_{ol}}{2R_m} \tag{22}$$

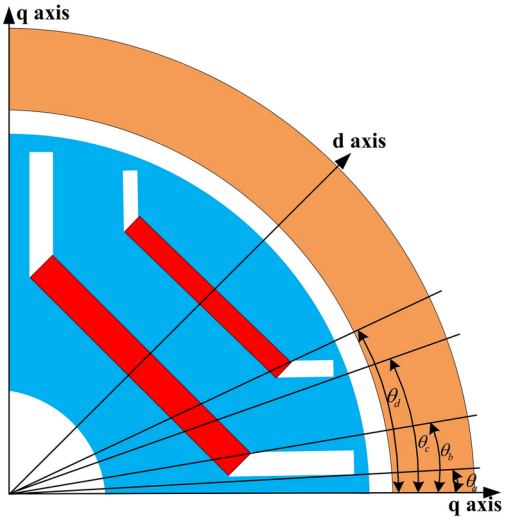
$$\phi_g = \frac{\frac{1}{2K_rR_g}}{\frac{1}{2K_rR_g} + \frac{1}{2R_{ib}} + \frac{1}{R_{ob}} + \frac{1}{2K_{ltotal}R_m}}\phi_{r1} \tag{23}$$

Analysis of no-load air gap flux density of double -layer rotor topologies

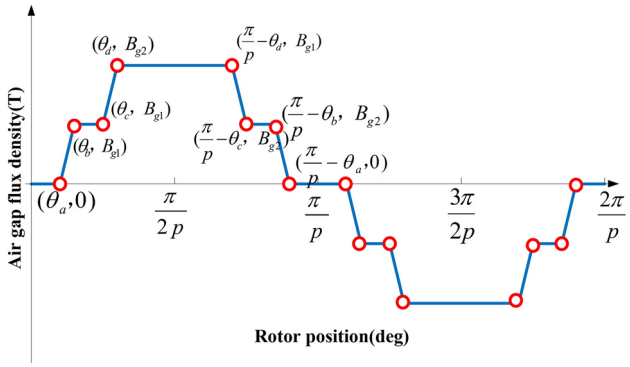
The air gap flux density distribution of slotless double-layer 2I type rotor topology structure is shown in Fig. 7. In the rotor of 2I type, the air gap region corresponding to permanent magnet should be divided into two parts.

In Fig. 7a, θ_a is the mechanical angle between q-axis and starting point of first layer of permanent magnet barrier, θ_b is the mechanical angle between q-axis and end point of first layer of permanent magnet barrier, θ_m is the mechanical angle between d-axis of permanent magnet and end point of magnetic barrier, θ_c is the mechanical angle between q-axis and starting point of second layer of permanent magnet barrier, and θ_d is the mechanical angle between q-axis and end point of second layer of permanent magnet barrier. The distribution of no-load flux density under individual excitation of permanent magnet with 2I type is shown in Eq. (24), and the corresponding waveform is shown in Fig. 7b. The Fourier transform of Eq. (24) yields Eq. (25).

$$B_{gn}(\theta) = \begin{cases} 00 \le \theta \le \theta_a \\ \dfrac{B_{g1}}{(\theta_b - \theta_a)}(\theta - \theta_a)\ \theta_a \le \theta \le \theta_b \\ B_{g1}\ \theta_b \le \theta \le \theta_c \\ \dfrac{B_{g1}}{(\theta_d - \theta_c)}(\theta - \theta_c)\ \theta_c \le \theta \le \theta_d \\ B_{g2}\ \theta_d \le \theta \le \dfrac{\pi}{p} - \theta_d \\ -\dfrac{B_{g2}}{(\theta_d - \theta_c)}\left(\theta - \left(\dfrac{\pi}{p} - \theta_c\right)\right)\ \dfrac{\pi}{p} - \theta_d \le \theta \le \dfrac{\pi}{p} - \theta_c \\ B_{g1}\ \dfrac{\pi}{p} - \theta_c \le \theta \le \dfrac{\pi}{p} - \theta_b \\ -\dfrac{B_{g1}}{(\theta_b - \theta_a)}\left(\theta - \left(\dfrac{\pi}{p} - \theta_a\right)\right)\ \dfrac{\pi}{p} - \theta_b \le \theta \le \dfrac{\pi}{p} - \theta_a \\ 0\ \dfrac{\pi}{p} - \theta_a \le \theta \le \dfrac{\pi}{p} \end{cases} \tag{24}$$



(a)Schematic diagram of 2I type



(b) Waveform of no-load air gap flux density

Fig. 7. Air gap flux density distribution of 2I type.

$$B_{gn} = \frac{4B_{g1}}{\pi N_p\left(\theta_b - \theta_a\right)(2n-1)^2}\left(\sin N_p\theta_b\,(2n-1) - \sin N_p\theta_a\,(2n-1)\right) + \frac{4\left(B_{g2} - B_{g1}\right)}{\pi N_p\left(\theta_d - \theta_c\right)(2n-1)^2}\left(\sin N_p\theta_d\,(2n-1) - \sin N_p\theta_c\,(2n-1)\right) \tag{25}$$

From Eqs. (24) and (25), it can be seen that when the amount of permanent magnet used is the same, compared with the single-layer structure, the air gap flux per pole decreases, but the sinusoidal degree of air gap flux density further increases, reducing the harmonic order.

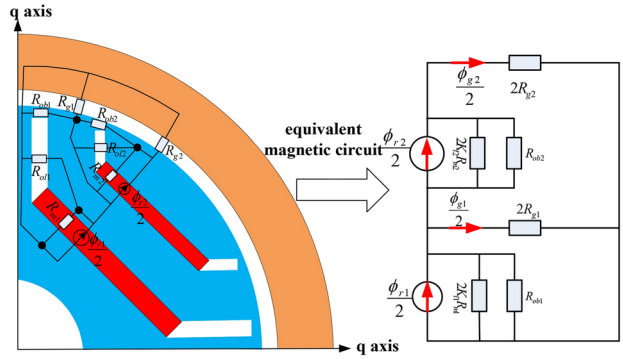
The double-layer rotor structure includes 2I type, 2 V type and IV type. When the number of permanent magnets, air gap area, and amount of permanent magnets in double-layer rotor structure are the same, the magnetic flux of each pole of rotor in the two regions varies according to the structural changes of the internal magnetic barrier and magnetic bridge. The equivalent magnetic circuit diagram of double-layer rotor topology structure is shown in Fig. 8. The air gap flux and air gap flux density of double-layer rotor topology structure can be calculated using the equivalent magnetic circuit in Fig. 8. The air gap flux and flux density of double layer rotor topology can be expressed as follows.

$$\phi_{g1} = \phi_{g11} - \phi_{g12} \tag{26}$$

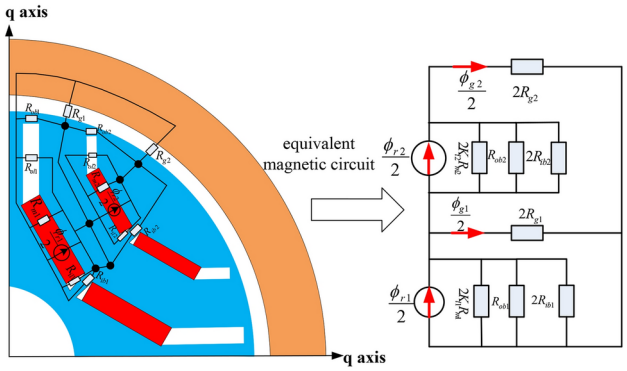
$$\phi_{g2} = \phi_{g21} + \phi_{g22} \tag{27}$$

$$B_{g1} = \frac{\phi_{g1}}{A_{g1}} \tag{28}$$

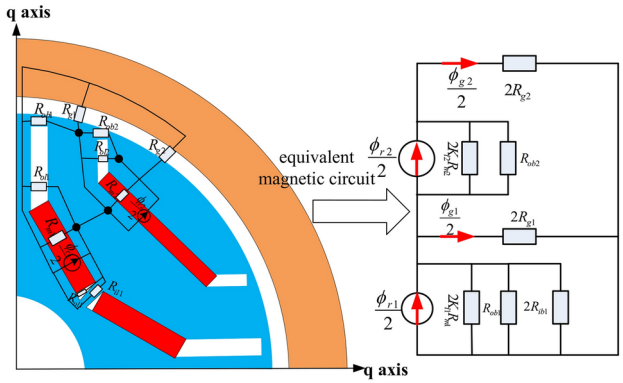
$$B_{g2} = \frac{\phi_{g2}}{A_{g2}} \tag{29}$$



(a) 2I type



(b) 2V type



(c) IV type

Fig. 8. Magnetic circuit of double rotor topology structures.

The air gap flux of 2I type can be expressed as

$$\phi_{g11} = \frac{P_{g1}}{P'_{m1} + P_{ob1} + P_{g1} + P_A}\phi_{r1} \tag{30}$$

$$\phi_{g21} = \frac{P_A}{P'_{m1} + P_{ob1} + P_{g1} + P_A}\phi_{r1} \tag{31}$$

$$\phi_{g12} = \frac{P_{g1}}{P'_{m1} + P_{ob1} + P_{g1} + P_A}\frac{P_B}{P'_{m2} + P_{ob2} + P_B}\phi_{r2} \tag{32}$$

$$\phi_{g22} = \frac{P_B}{P'_{m2} + P_{ob2} + P_B}\phi_{r2} \tag{33}$$

The air gap flux of 2 V type can be expressed as

$$\phi_{g11} = \frac{P_{g1}}{P'_{m1} + P_{ob1} + P_{ib1} + P_{g1} + P_A}\phi_{r1} \tag{34}$$

$$\phi_{g21} = \frac{P_A}{P'_{m1} + P_{ob1} + P_{ib1} + P_{g1} + P_A}\phi_{r1} \tag{35}$$

$$\phi_{g12} = \frac{P_{g1}}{P'_{m1} + P_{ob1} + P_{ib1} + P_{g1}} \frac{P_B}{P'_{m2} + P_{ob2} + P_B}\phi_{r2} \tag{36}$$

$$\phi_{g22} = \frac{P_B}{P'_{m2} + P_{ob} + P_{ib} + P_B}\phi_{r2} \tag{37}$$

$$P_A = 2K_{l2}R_{m2} \| R_{ob2} \| R_{ib2} + 2R_{g2} \tag{38}$$

$$P_B = 2K_{l1}R_{m1} \| R_{ob1} \| R_{ib1} + 2R_{g2} \tag{39}$$

The air gap flux of IV type can be expressed as

$$\phi_{g11} = \frac{P_{g1}}{P'_{m1} + P_{ob1} + P_{ib1} + P_{g1} + P_A}\phi_{r1} \tag{40}$$

$$\phi_{g21} = \frac{P_A}{P'_{m1} + P_{ob1} + P_{ib1} + P_{g1} + P_A}\phi_{r1} \tag{41}$$

$$\phi_{g12} = \frac{P_{g1}}{P'_{m1} + P_{ob1} + P_{ib1} + P_{g1}} \frac{P_B}{P'_{m2} + P_{ob} + P_D}\phi_{r2} \tag{42}$$

$$\phi_{g22} = \frac{P_B}{P'_{m2} + P_{ob} + P_B}\phi_{r2} \tag{43}$$

$$P_A = 2K_{l2}R_{m2} \| R_{ob2} + 2R_{g2} \tag{44}$$

$$P_B = 2K_{l1}R_{m1} \| R_{ob1} \| 2R_{g1} + 2R_{g2} \tag{45}$$

The analyses of the above three double-layer rotor topologies show that due to the lack of internal magnetic bridges and barriers causing leakage flux in 2I type, the air gap flux density of 2I type is the highest, while the 2 V type has the lowest air gap flux density. The flux of each pole of IV type is between 2I and 2 V type, but due to the absence of magnetic barriers and bridges between the two layers of permanent magnets, the internal flux of IV type is larger and the external flux is smaller compared to the double-layer 2 V type.

Based on the above analysis, it can be found that the I type structure has good suppression ability for magnetic leakage, but the permanent magnet does not have a magnetic focusing effect, and the circuit of cross axis has a single magnetic channel, which is not conducive to improving the reluctance torque. The V type structure has a magnetic focusing effect, but there is severe magnetic leakage, while the IV type structure has a more pronounced magnetic focusing effect, and the magnetic leakage is constrained.

The double-layer permanent magnet structure can improve the sinuosity of air gap flux density, especially the double-layer IV type permanent magnet structure. By adding an I type structure on the basis of V type structure, the permanent magnetomotive force can be adjusted to a stepped wave with a middle height and slightly lower shoulders, resulting in the highest sinuosity of air gap flux density. The multi-layer permanent magnet magnetic barrier further hinders the d-axis magnetic circuit, while the q-axis magnetic circuit is minimally affected. Therefore, it can increase the salient pole ratio, which is beneficial for improving the reluctance torque and reducing the amount of permanent magnets used. However, it should be noted that having more layers of permanent magnets does not necessarily mean better. Under the same constraint of permanent magnet usage, the more layers there are, the more complex the topology structure of the permanent magnet rotor, the thinner the permanent magnet, and the more complex the processing technology of the rotor and permanent magnet.

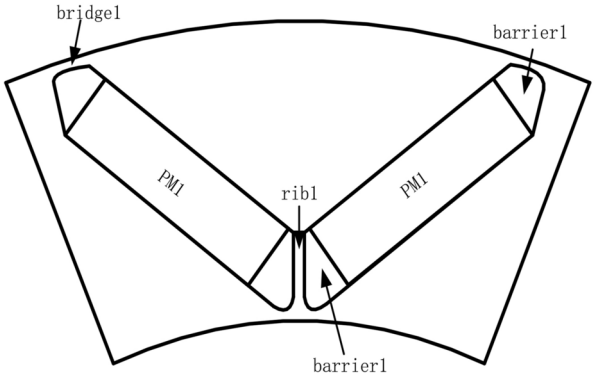
In order to further improve sinuosity of waveform of air gap flux density and utilization rate of reluctance torque, this paper proposes three-layer rotor topology structure with I2V type and a parameterized finite element analysis (FEA) model is proposed to optimize the motor. The aim is to achieve low iron consumption, high efficiency, and high torque density while realizing low torque ripple in IPMSM for vehicles, providing a certain analytical reference for the optimization design of motors for electric vehicle applications.

The above analysis mainly uses magnetic circuit equivalent method, namely lumped constant magnetic circuit method, to analyze and compare the magnetic field distribution in air gap of different rotor structures. The magnetic circuit equivalent circuit method is fast and intuitive, and is useful for simple design and prediction of motors. However, the magnetic circuit equivalent method does not consider issues such as cogging effect, leakage magnetic flux, and magnetic saturation of the isolation bridge, and cannot accurately obtain the magnetic field distribution of IPMSMs. The finite element method is a commonly used modeling method in motor design, which is a distributed constant magnetic circuit method that can accurately analyze the effects of torque ripple, cogging torque, and voltage harmonic components on spatial and temporal magnetic field changes.

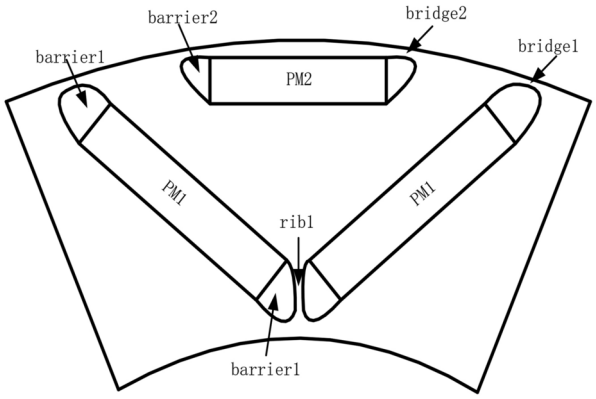
Modeling and optimization of rotor topology structure

Modeling of rotor topology structures

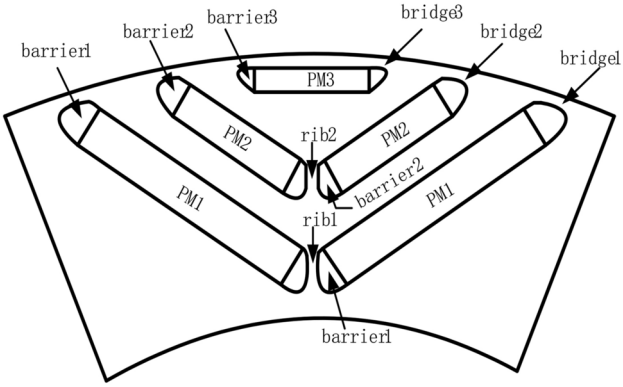
Three different rotor topology structures are shown in Fig. 9. In order to ensure comparability of analysis results, the same stator structure is used for three different rotor topologies of the IPMSMs. The rotor structure mainly consists of rotor iron core, magnetic barriers corresponding to each layer of permanent magnets, magnetic bridges separating each layer, ribs strengthening the rotor structure between the permanent magnet magnetic barriers and permanent magnets.



(a) V type



(b) IV type



(c) I2V type

Fig. 9. Different rotor topology structures.

Design parameters	Value
Rated power	70 kW
Peak power	130 kW
Rated speed	4500r/min
Maximum speed	16000r/min
DC voltage	600 V
Phase poles	3phases 8poles
Number of stator teeth	48(distributed)
Stator materials Rotor materials	M250-35A M250-35A
N38UH(20℃)	Br = 1.26 T,Hc = -954930A/m
Diameter of stator	220 mm
Diameter of rotor	142.5 mm
The length of air gap	0.7 mm
Continuous phase current amplitude	300A
Maximum phase current amplitude	500A

Table 1. Characteristics of IPMSMs.

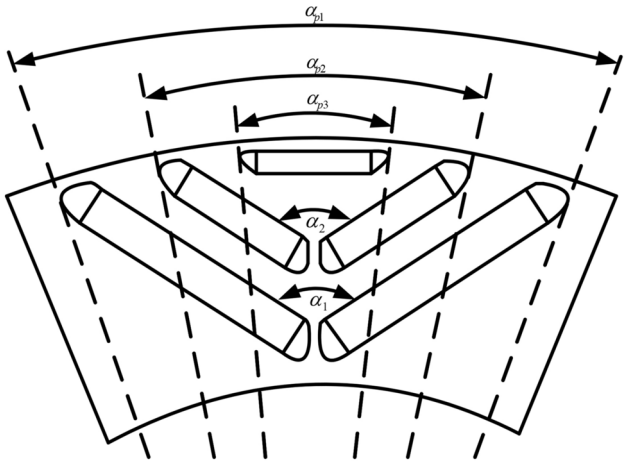


Fig. 10. Schematic of rotor topology parameters.

The parameters of IPMSM are shown in Table 1. The same stator and windings of three motors is adopted, the stator poles and slots is 8 poles and 48 slots, and the stator adopts a full distance distributed winding. Driven by a three-phase sine current source, with a rated current amplitude of 260A and a maximum peak current amplitude of 500A, using liquid cooling.

Optimization of rotor topology structures

The rotor topology structure has a significant impact on the d-q inductance L_d and L_q, and no-load air gap flux density waveform of IPMSM. As the d-q inductance and harmonic content of no-load air gap flux density waveform has a significant impact on torque density, torque ripple, stator iron loss, and reluctance torque utilization rate. This section mainly conducts optimization analysis on the two important parameters that have the greatest impact on the electromagnetic performance of the IPMSM, namely the opening angle of the first layer V type permanent magnet and the pole arc angle of the first layer V type permanent magnet, through the established 2D FEA parameterized models. Based on this, the rotor topology structure is further optimized and determined.

The optimization of PM1 opening angle

The main parameters of three-layer rotor topology structure are shown in Fig. 10. In Fig. 10, α_1 and α_2 represent the mechanical angles at which the first and second layers of V type permanent magnets are opened, respectively. α_{p1}, α_{p2}, and α_{p3} represent the pole arc angles of the first, second, and third layers of permanent magnets (PMs), respectively.

The effect on motor performance of opening mechanical angle of first layer V type permanent magnet (PM1) is shown in Fig. 11. From Fig. 11, it can be observed that the torque density increases with increase of opening

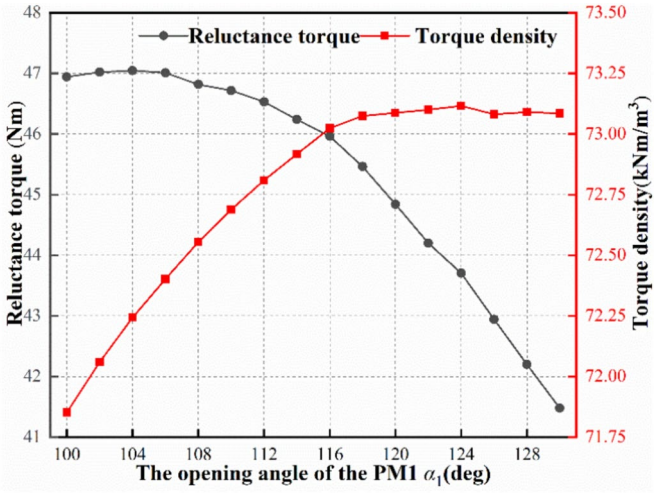


Fig. 11. Influence on performance of PM1 Angle.

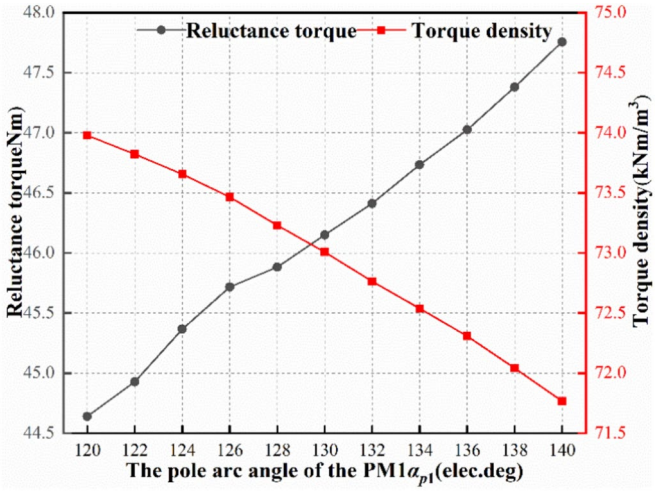


Fig. 12. Influence on performance of PM1 pole-arc angle.

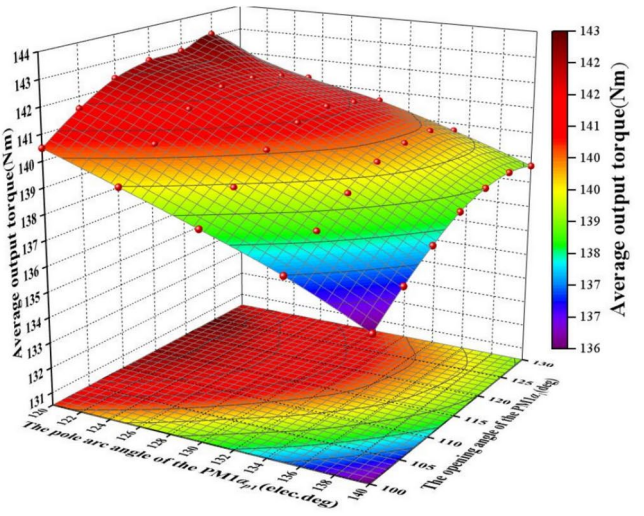
angle of first layer of V type permanent magnet. When the value of α_1 is around 115°, the torque density increases slowly and remains basically unchanged. The reluctance torque decreases with the increase of opening angle of first layer of V type permanent magnet. When α_1 is set to 115°, both torque density and reluctance torque can be simultaneously considered.

The optimization of PM1 pole arc angle

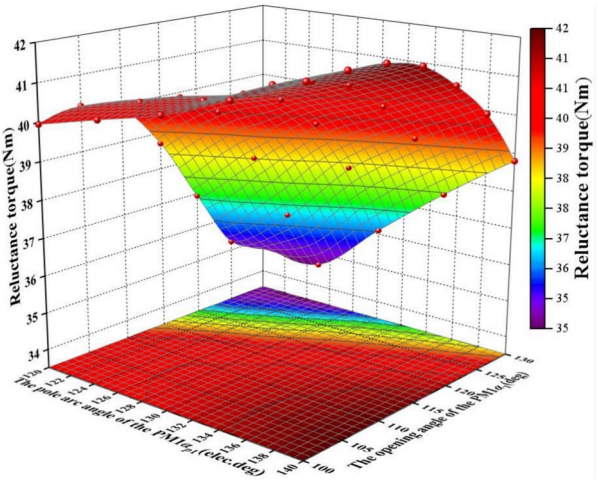
The influence on motor performance of the first layer V type permanent magnet pole arc angle is shown in Fig. 12. From Fig. 12, it can be observed that the reluctance torque increases with the increase of pole arc angle, while the torque density decreases with the increase of pole arc angle. When the pole arc angle is set to 130°, both the torque density and reluctance torque output can be simultaneously considered.

The relationship between torque output and the opening angle as well as the pole arc angle of PM1 is shown in Fig. 13. It can be observed from Fig. 13 that when the opening angle of PM1 is set to 115° and the pole arc angle of PM1 is set to 130°, the average output torque and reluctance torque of IPMSM can both reach their maximum values, which not only improves the torque density of the motor, but also fully utilizes the reluctance torque.

Under the constraint of equal permanent magnet usage, the final optimized parameter values of three different topology structures are shown in Table 2. Where, PMT, PMW and PWV are the thickness, width, and total mass of the permanent magnet for each rotor topology, respectively.



(a) Average output torque

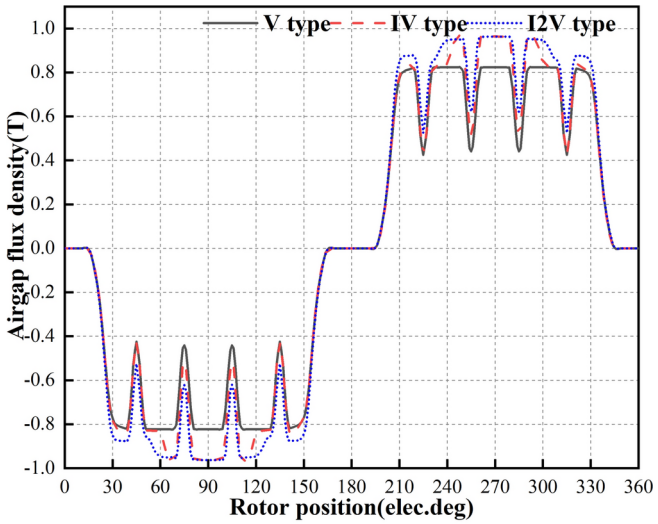


(b) Reluctance torque

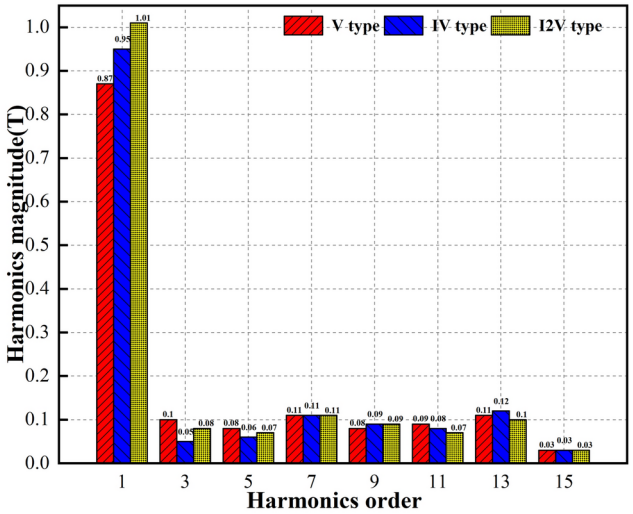
Fig. 13. Output torque according of the PM1 angle and pole-arc Angle.

Parameters	V type	IV type	I2V type
α_1(deg)	115	115	115
α_2(deg)	NAN	NAN	115
$\alpha_{p1}(elec.\ \mathrm{deg})$	130	130	130
$\alpha_{p2}(elec.\ \mathrm{deg})$	NAN	66	78
$\alpha_{p3}(elec.\ \mathrm{deg})$	NAN	NAN	50
PMT1 (mm)	7	4.5	3.5
PMT2 (mm)	NAN	4	3
PMT3 (mm)	NAN	NAN	2
PMW1 (mm)	18	20	22
PMW2 (mm)	NAN	18	12
PMW3 (mm)	NAN	NAN	12
PMV (kg)	2.011	2.011	1.995

Table 2. Optimal values of parameters with three rotors.



(a) Comparison of waveform distribution



(b) Comparison of Harmonic Distribution

Fig. 14. Comparison of no-load air gap flux density.

Parameters	V type	IV type	I2V type
Fundamental values (T)	0.95	0.92	0.95
Harmonic distortion rate (%)	24.6	20.5	17.9

Table 3. Amplitude of the fundamental and THD of no-load air-gap flux density.

The results comparison of FEA

Analysis and comparison of no-load air gap flux density

The waveform and harmonic distribution of no-load air gap flux density of the optimized rotor topologies are compared in Fig. 14. From Fig. 14a, it can be seen that compared with the other two rotors, the IPMSM with I2V type has the highest sinusoidal waveform and the lowest total harmonic distortion rate.

According to Fig. 14b, the fundamental values and total harmonic distortion rates of no-load air gap flux density with three rotor topologies are shown in Table 3. From Table 3, it can be seen that the fundamental amplitude of no-load air gap flux density with IV type is the smallest, while the fundamental amplitude of V type and I2V type is the largest. The total harmonic distortion rate of V type is the highest, the I2V type is the smallest, and compared with the V type, the total harmonic distortion rate of I2V type is reduced by 27.2%.

Analysis and comparison of torque performance

Torque comparison under different current densities

The average output torque versus current density with different current angles is shown in Fig. 15.

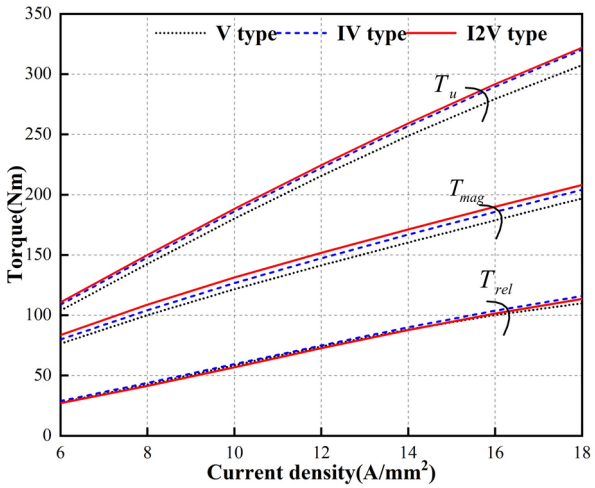
From Fig. 15, it can be observed that the total average output torque T_u increases with increase of current density. The total average output torque of IPMSM with I2V rotor topology is greater than the other two at different current densities. As shown in Fig. 15a, under flux weakening conditions with current angle of 30°, The permanent magnet torque T_{mag} of IPMSM with I2V is significantly higher than the other two. As shown in Fig. 15b, under deep flux weakening conditions with a current angle of 70°, The reluctance torque T_{rel} of IPMSM with I2V is significantly higher than that of the other two.

The comparison of torque density under different current densities with three different rotor topologies is shown in Fig. 16. From Fig. 16, it can be seen that under different current densities, the torque density of IPMSM with I2V rotor topology is higher than the other two. Compared with V type, as the current density increases, the difference in torque density also increases. When the current density is 8A/mm^2, the minimum difference of torque density is 3.6 kNm/m^3, and when the current density is 18A/mm^2, the maximum difference of torque density is 7.1 kNm/m^3. Compared with V type, the IPMSM with I2V rotor topology has an increase in torque density of about 5.0% with slightly lower permanent magnet usage.

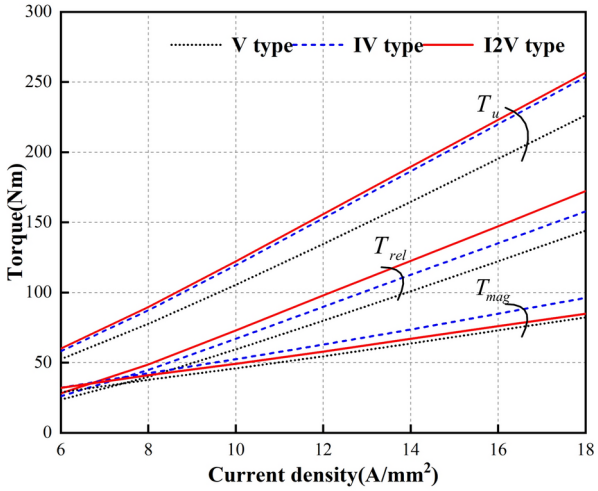
Analysis and comparison of torque waveform

The output torque waveforms of IPMSMs with three different rotor topologies under rated operating conditions at speed of 4500 rpm, current amplitude of 260A, and current angle of 30° are shown in Fig. 17. From Fig. 17, it can be seen that I2V type has the maximum output torque, while the V type has the minimum output torque. IV type has the smallest torque ripple, while I2V type has the largest torque ripple.

The comparison of average torque and torque ripple with three different rotor topologies under rated operating conditions is shown in Table 4. In Table 4, T_{av} represents average output torque and T_{rip} represents torque ripple.



(a) Current angle 30°



(b) Current angle 70°

Fig. 15. Average output torque versus current density with different current angles.

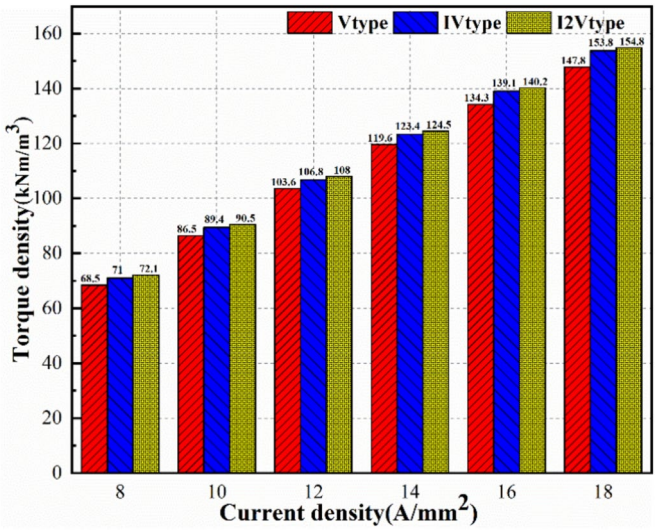


Fig. 16. Torque density versus current density.

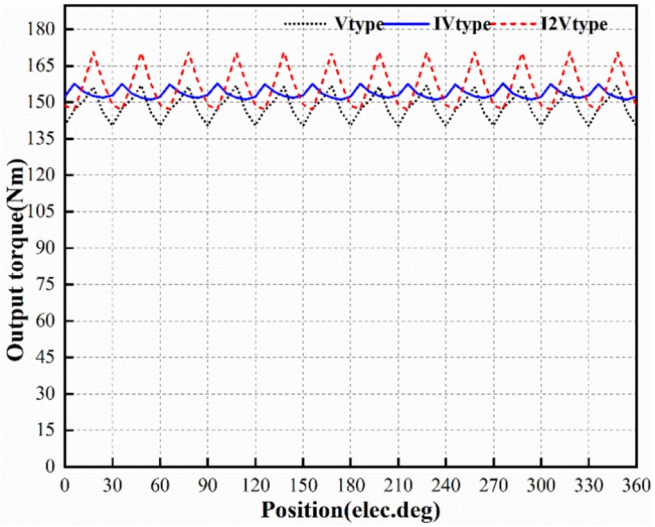


Fig. 17. Comparison of output torque waveforms of different rotor topology structures.

Parameters	V type	IV type	I2V type
T_{av} (Nm)	148.3	153.8	156.8
T_{rip} (Nm)	16.4	7.0	24.5
T_{rip} (%)	11.0	4.6	15.6

Table 4. Comparison of average torque and torque ripple.

It can be clearly seen from Table 4 that the I2V type has the highest average output torque. Compared with V type, the average output torque of I2V type has increased by 8.5 Nm, with an increase of 5.7%. Compared with IV type, the average output torque of I2V type has increased by 3Nm, with an increase of 2.0%. However, the I2V type rotor topology has the highest torque ripple and corresponding ripple ratio, which are 24.5Nm and 15.6%, respectively.

In summary, under rated operating conditions, the average output torque of I2V type is greater than that of the other two, and the torque ripple of I2V type is also greater than that of the other two. Therefore, when the motor is connected to the transmission system, it will significantly increase the vibration and noise of the transmission system. Therefore, it is necessary to further optimize rotor topology structure of I2V type to reduce its torque ripple effectively.

Optimization of torque ripple with I2V type

Optimizing the topology structure of the IPMSM can reduce torque ripple, and rotor skewed poles can effectively reduce torque ripple[12,13]. Asymmetric permanent magnet structure[14], adding magnetic barriers[15], rotor pole displacement[16], adding auxiliary permanent magnets[17], changing the shape or segmentation of the permanent magnets[18]. These methods have been proven to effectively reduce torque ripple under specific conditions, but due to factors such as processing costs, technological maturity, and electromagnetic performance degradation, they have not yet been widely applied in the optimization design of IPMSMs.

From the perspective of the equivalent magnetic circuit, the magnetic circuit of IPMSM is mainly composed of air gap, PM, teeth, and yoke. The torque ripple of IPMSM is mainly predominantly attributed to the nonlinearity of magnetic circuit and the uneven distribution of PMs. The introduction of magnetic isolation holes can improve the magnetic field distribution between these components and optimize the magnetic circuit structure, which can effectively reduce the torque ripple of IPMSM. This principle is primarily based on the following considerations:

(a) Magnetic isolation hole effectively divides the magnetic circuit into multiple small magnetic circuits by introducing non-magnetic materials into the rotor magnetic bridge section, which can increase the magnetic resistance in the magnetic flux path. As the magnetic permeability of non-magnetic materials is much lower than that of ferromagnetic materials, the increased magnetic resistance makes the distribution of magnetic flux more uniform, reduces fluctuations in local magnetic flux density, and thus reduces torque ripple.
(b) The presence of magnetic isolation hole makes the magnetic flux more evenly distributed in each small magnetic circuit. Due to the uniform distribution of magnetic flux, the torque generated by the IPMSM during operation is also more stable, reducing the periodic fluctuations of torque, namely torque ripple.
(c) In IPMSMs, high magnetic saturation at the magnetic bridge can cause uneven magnetic flux distribution, leading to torque ripple. The magnetic isolation hole reduces the flux density of magnetic bridge by increasing magnetic resistance, thereby avoiding over saturation of the magnetic isolation bridge, making magnetic flux distribution more linear, then reducing torque ripple.
(d) Optimization design of magnetic isolation hole can suppress harmonic magnetic fields generated inside the IPMSM. Harmonic magnetic field is an important factor causing torque ripple. By optimizing the design of magnetic isolation holes, the generation of these harmonic magnetic fields can be reduced, thereby further reducing torque ripple.
(e) In the model of equivalent magnetic circuit, the introduction of magnetic isolation hole changes magnetic circuit structure, making the equivalent magnetic circuit model more complex. But this complexity is balanced by increasing magnetic resistance, making magnetic flux distribution calculated by the model more accurate, which is reflected in the reduction of torque ripple.
(f) Optimization design of magnetic isolation hole not only reduces torque ripple, but also improves the efficiency of IPMSM. By reducing the saturation of magnetic bridge, the hysteresis loss and eddy current loss are lowered, making the IPMSM more efficient during operation.

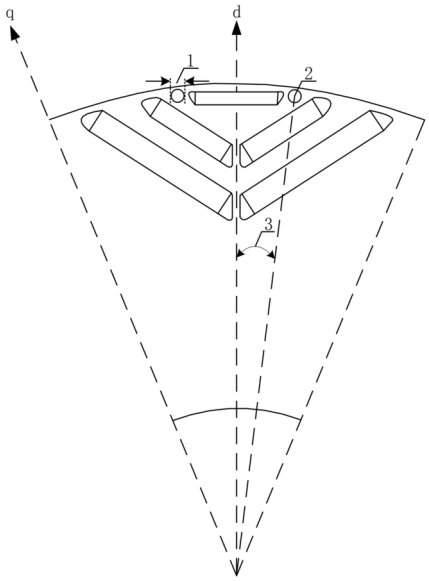
In summary, by designing the position and size of magnetic isolation hole reasonably, the direction of magnetic flux lines can be controlled to avoid excessive concentration in one area. In other words, the magnetic isolation hole can optimize the magnetic circuit structure, improve the magnetic flux distribution, reduce the saturation phenomenon of the magnetic isolation bridge, and effectively reduce the torque ripple of IPMSM. So it is necessary to establish a parameterized model of magnetic isolation hole to optimize and adjust the position and diameter of magnetic isolation hole for each pole, effectively reducing torque ripple, hysteresis loss, and eddy current loss. This optimized design not only improves the torque performance, but also reduces hysteresis and eddy current losses, improving the operating efficiency.

Regarding the issue of maximum torque ripple of IPMSM with I2V type among three rotor topologies, a simple and effective method of adding a set of circular magnetic isolation holes to each pole of the rotor is proposed. By establishing a parameterized model of circular magnetic isolation holes and optimizing the position and diameter of the set of circular magnetic isolation holes, the torque ripple of I2V type can be effectively reduced. The parameterized optimization model of rotor with I2V type is shown in Fig. 18.

In order to achieve both minimum torque ripple and maximum output average torque simultaneously, as shown in Fig. 18, a set of circular magnetic isolation holes are added between the second and third layers of permanent magnets at each pole of rotor. By establishing a parameterized analysis model for circular magnetic isolation holes, the position of circular magnetic isolation hole, namely the distance between center of circular magnetic isolation hole and rotor center, and the angle between center of circular magnetic isolation hole and the d-axis, were optimized and adjusted. The optimization process of torque ripple is shown in Fig. 19. After parameter optimization and adjustment, the optimized value of the distance between center of magnetic isolation hole and rotor center is determined to be 69.25 mm, the optimized value of angle θ_h between center of magnetic isolation hole and the d-axis is 7°, and diameter range of magnetic isolation hole is 1.5 mm ~ 2 mm.

The objective function is to minimize torque ripple and maximize output torque. The constraint conditions are set to prevent a decrease in output torque and an increase in torque ripple during the optimization process. The optimization objective is to minimize torque ripple T_{rip} and maximize average output torque T_{av}. The constraints are that the current amplitude is 260A, the current angle is 30°, and the continuous average output torque T_{av} is not less than 150 Nm, the torque ripple ratio T_{rip} (%) is not greater than 6%, and the optimization variable value range is $1.5\ \text{mm} \le d_h \le 2\ \text{mm}$.

The optimization process is based on initial FEA results of IPMSM with I2V type. Due to the torque ripple ratio being 15.0% of initial value, we hope that the optimized result will have a much smaller torque ripple ratio than initial value while keeping average output torque basically unchanged.



1. Diameter of the magnetic isolation hole d_h 2. Position radius of the magnetic isolation hole r_h 3. Angle between the center of the magnetic hole and the d-axis θ_h

Fig. 18. Schematic diagram of position and structure of magnetic isolation hole with one pole.

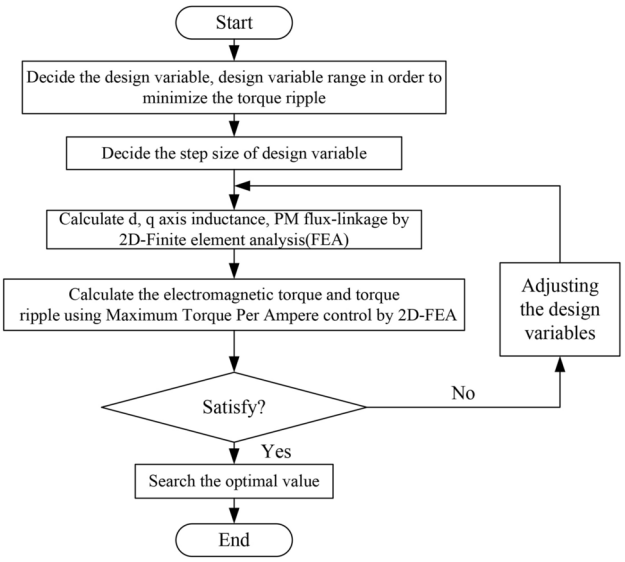


Fig. 19. Optimization procedure flow chart with I2V type.

Under rated operating conditions, the relationship between output torque waveform of IPMSM and the diameter of magnetic isolation hole is shown in Fig. 20. From Fig. 20, it can be observed that under the condition of minimal topological interference, the torque ripple of I2V type decreases as the diameter of magnetic isolation hole increases. When final optimized value of the diameter of magnetic isolation hole is 2 mm, the torque ripple and ripple ratio of IPMSM are all minimized.

Under rated operating conditions, the comparison of output torque waveforms of I2V type after optimization is shown in Fig. 21. From Fig. 21, it can be seen that the optimized torque ripple of I2V type has significantly reduced. Compared with the torque ripple value without optimization, the optimized torque ripple value is 8.0Nm, a significant decrease of 16.5Nm, with a reduction ratio of 67.3%,the optimization effect is very significant.

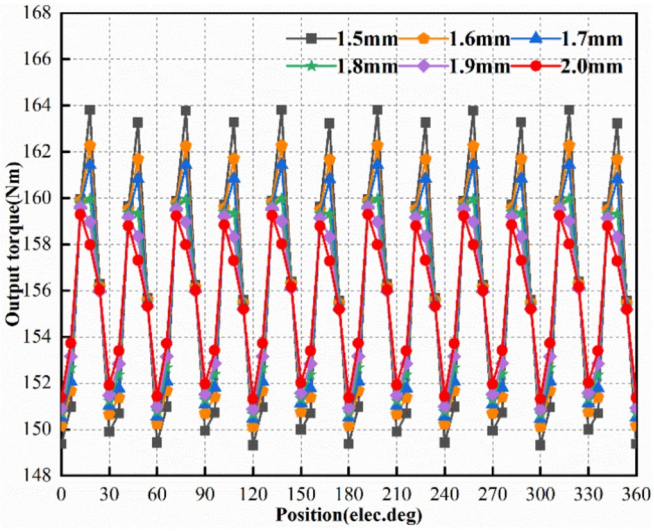


Fig. 20. Comparison of torque waveforms under different flux isolation hole diameters.

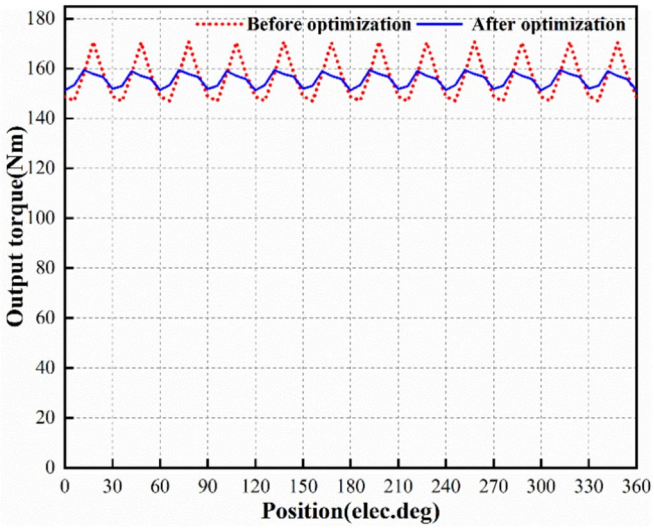


Fig. 21. Comparison of torque waveforms after optimization.

The field magnetic field distributions

The FEA results of the magnetic field and flux density under rated operating conditions of three different rotor topologies are shown in Figs. 22 and 23, respectively. From Figs. 22 and 23, it can be seen that the magnetic field distribution of the IPMSM with different rotor topologies is reasonable under rated operating conditions.

The loss and efficiency maps

During the driving process of electric vehicles, they may experience various complex operating conditions, such as traffic, low-speed climbing, urban operating conditions, high-speed operating conditions, etc. Therefore, their operating characteristics determine that the drive motor of electric vehicle must have a wide speed range and the ability to maintain high efficiency operation within a wide speed range. So in addition to torque performance in the design of drive motors, the losses and efficiency are also key indicators.

The total loss maps

The losses of IPMSM mainly include copper losses, iron losses, and PM eddy losses. The total loss map of IPMSM with three different rotor topologies is shown in Fig. 24. From Fig. 24, it can be clearly seen that the total loss of IPMSM increases with the increase of torque and speed, and the overall high loss area is mainly concentrated in small area with deep flux weakening condition and high torque output. Under the same speed and torque of IPMSM with three different rotor topologies, the I2V type rotor topology proposed in this paper has the lowest total loss and the advantages are more obvious.

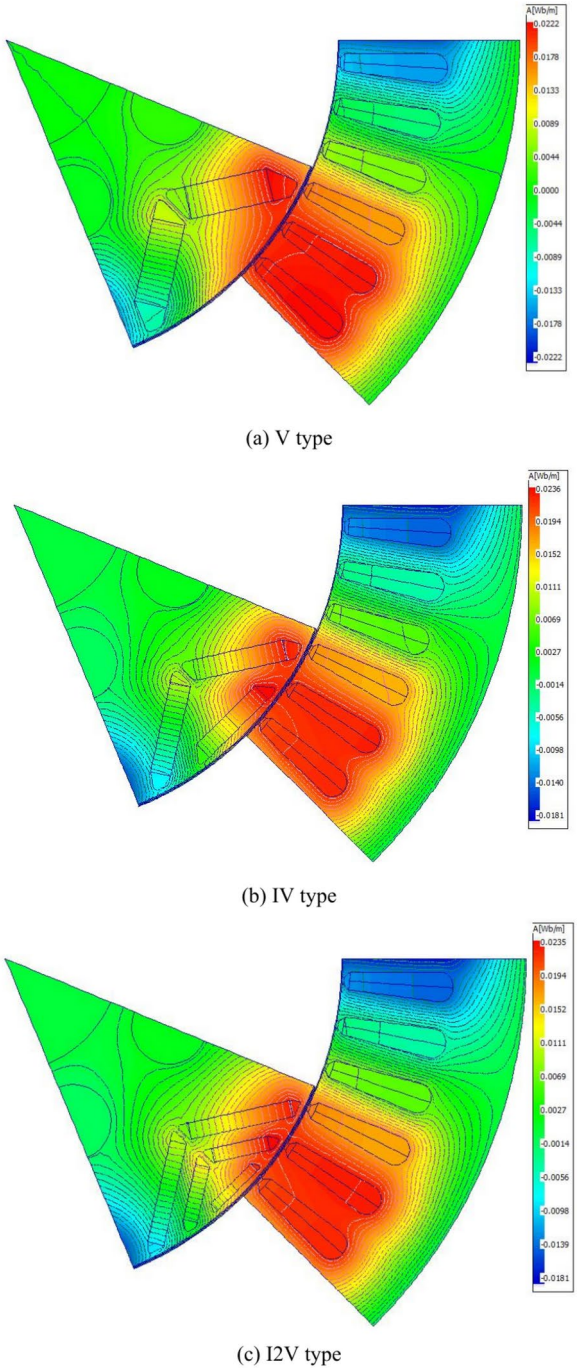


Fig. 22. Field distribution of IPMSM with different rotor topologies under rated operating conditions.

The efficiency maps

The efficiency maps of three types are shown in Fig. 25. From Fig. 25, it can be seen that the high-efficiency operating area of IPMSMs with I2V type is significantly larger than that of the other two. The comparison of the proportion of high-efficiency areas among the three types is shown in Table 5. From Table 5, it can be seen that I2V type has the highest proportion of high-efficiency areas. Compared with the V type, the proportion of areas with efficiency greater than 97% in the I2V type has increased by 6.32 percentage points, and the proportion of areas with efficiency greater than 96% has increased by 2.95 percentage points.

Prototype manufacturing and experimental verification

Strength analysis of rotor structure with I2V type

It has become a feasible way to use the FEA method to analyze and check the structural strength of the motor[19]. As the speed and rotor temperature of IPMSM with high torque density designed in this paper are relatively

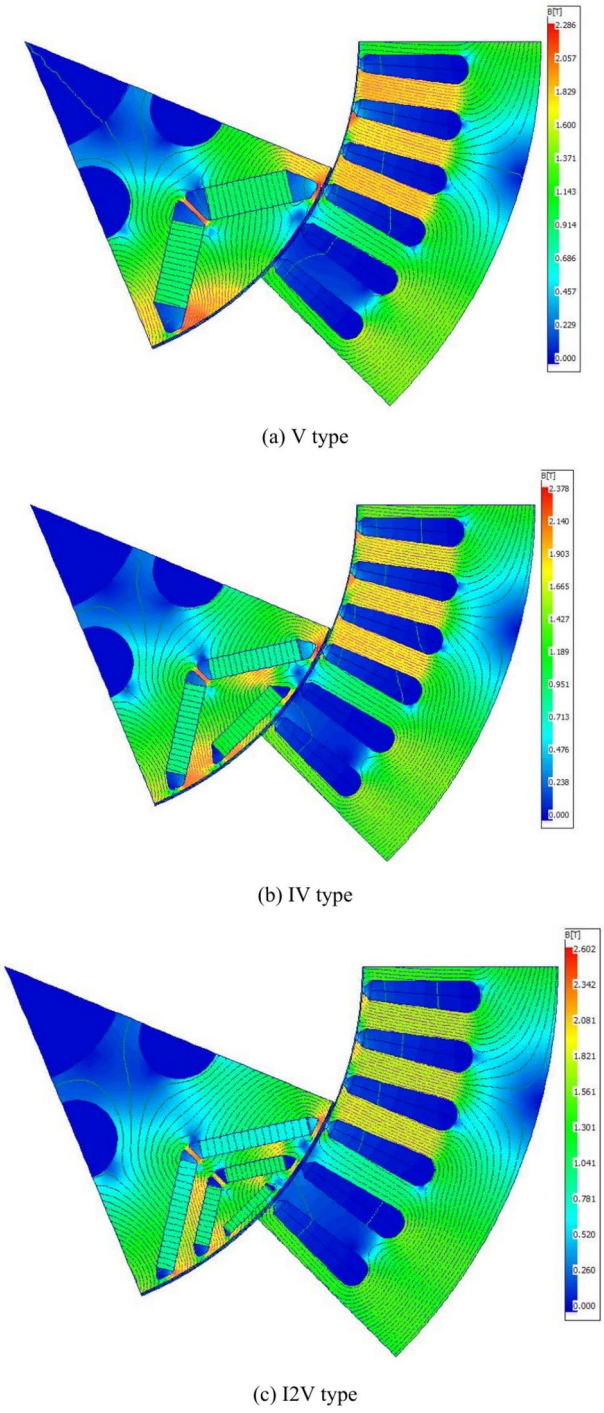
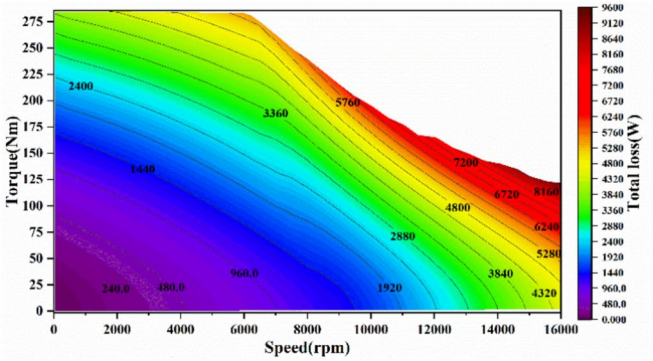


Fig. 23. Flux density of IPMSM with different rotor topologies.

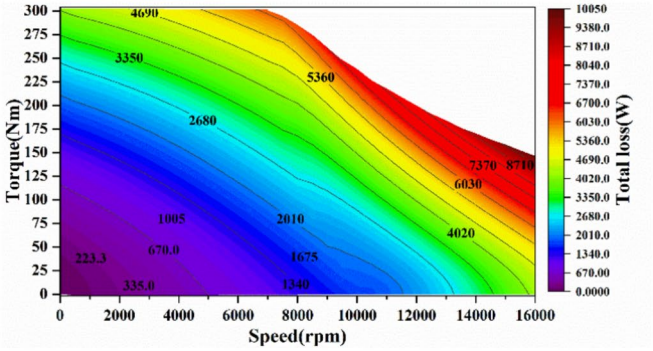
high. Therefore, the rotor structural strength needs to be focused on and calculated. According to the symmetry of the motor, the 1/8 rotor model is used for FEA and calculation. The material properties[20,21] of rotor are shown in Table 6.

The rotor stress and deformation nephogram at the maximum speed of 18000 rpm is shown in Fig. 26. Figure 26a shows the FEA Von-Mises stress distribution nephogram of the rotor at the maximum speed of 18000 rpm. It can be seen from the Fig. 26a that the maximum mechanical stress intensity point is located at rib1, with the size of 426.92Mpa. The FEA Von-Mises stress value of the rotor does not exceed the maximum yield strength of silicon steel sheet with 455Mpa. Therefore, when the motor runs at the highest speed, the rotor will not rupture.

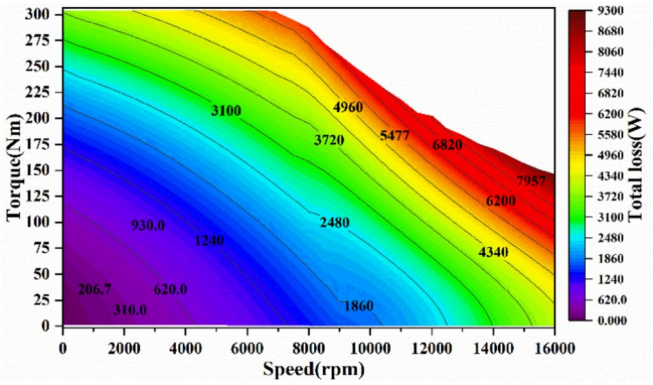
Figure 26b shows the distribution nephogram of the mechanical deformation of the rotor at the maximum speed of 18000 rpm. From Fig. 26b, it can be seen that the maximum mechanical deformation displacement



(a) V type



(b) IV type



(c) I2V type

Fig. 24. Total loss maps of IPMSM with three different rotor topologies.

point is located at bridge3, with the size of 0.0555 mm. the maximum displacement value is far less than air gap length, which will not cause contact and friction between stator and rotor.

Prototype manufacturing of I2V type

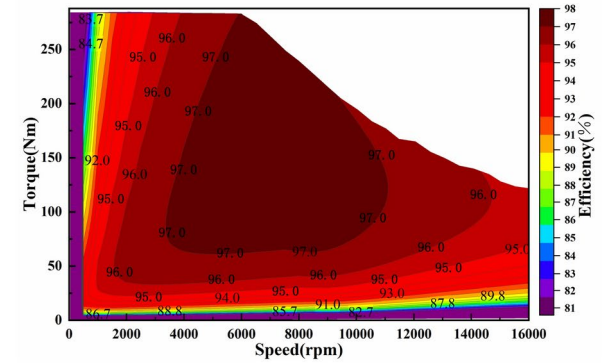
The rotor lamination, stator and rotor structure of I2V prototype are shown in Fig. 27.

The rotor and stator of I2V prototype are shown in Fig. 28, and the assembly diagram of rotor and shaft is shown in Fig. 28(a). Permanent magnet model embedded in the rotor is N38UH, and rotor position sensor adopts a four pair pole reluctance rotary transformer, model J52XU9734A.

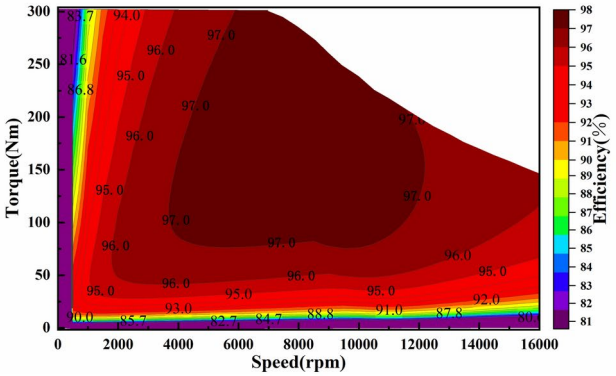
The stator winding is shown in Fig. 28b. The stator winding adopts double-layer full distance winding, with a span of 6 slots and 4 parallel connections. The end of the winding adopts star connection.

Prototype performance testing of I2V type

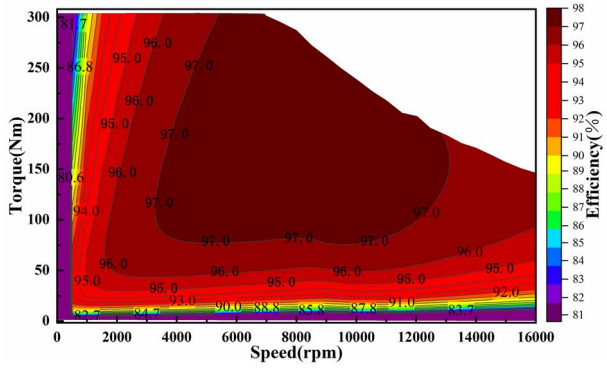
The prototype experimental test bench is shown in Fig. 29. The motor test adopts the Austrian List (AVL) test bench, including DynoSpirit dynamometer, 800 V battery simulator, speed sensor and its measuring instrument, torque sensor and its measuring instrument, and liquid cooled temperature control system.



(a) V type



(b) IV type



(c) I2V type

Fig. 25. Efficiency maps of IPMSM with three different rotor topologies.

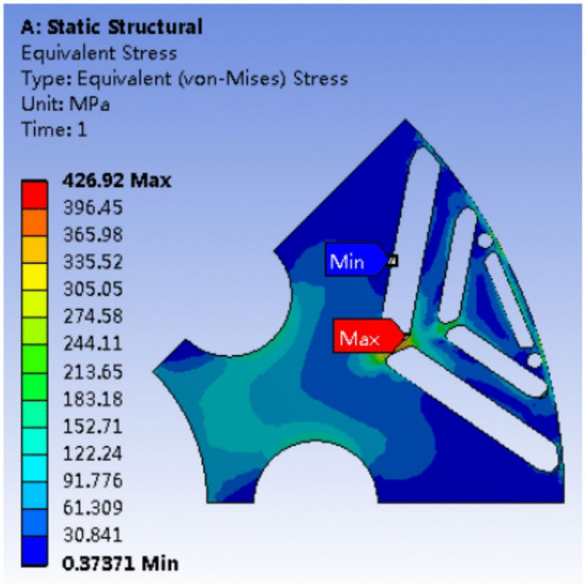
Parameter	V type	IV type	I2V type
Efficiency greater than 97%	33.08%	34.80%	39.40%
Efficiency greater than 96%	60.58%	61.78%	63.53%
Efficiency greater than 95%	72.94%	72.55%	73.59%

Table 5. Comparison of proportion of high efficiency area.

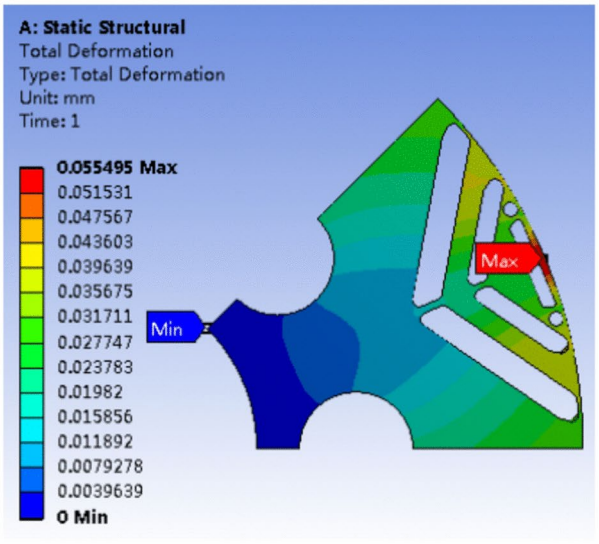
The prototype to be tested is connected to the dynamometer through a coupling. The prototype inverter driver can automatically adjust the proportional integral control parameters of speed loop and current loop in the vector control. The inverter driver pre writes a parameter table based on simulation results and uses a lookup table method to perform flux weakening speed regulation control on the IPMSM, obtaining experimental results at different speeds and loads of the prototype.

Parameters	Steel
Young's modulus (GPa)	185
Density (kg/m^3)	7650
Poisson's ratio	0.3
Yield strength (MPa)	455

Table 6. The material properties of rotor.



(a) The FEA results of rotor stress



(b) The FEA results of rotor deformation

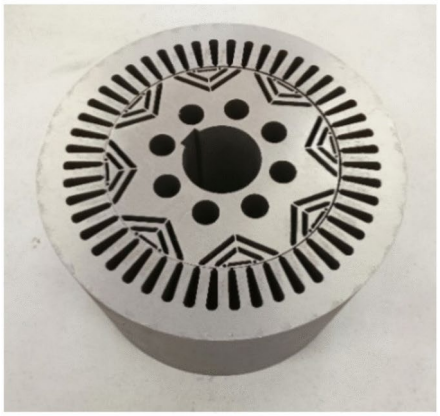
Fig. 26. The rotor stress and deformation nephogram at the maximum speed of 18000 rpm.

Prototype torque performance testing

The prototype drive controller is controlled by max torque per ampere (MTPA), and the dynamometer adopts speed control mode. When the current angle is 30°, the FEA and actual measurement results of output torque under different drive currents are compared as shown in Fig. 30. From Fig. 30, it can be observed that the torque simulation and actual measurement results are basically consistent. When the current amplitude is rated at 260A, the measured result is 3.3% smaller than the FEA result. When the current amplitude is 380A, the



(a) Rotor lamination



(b) stator and rotor structure

Fig. 27. Structure of rotor and stator.

measured result is about 4.2% smaller than the FEA result; When the current amplitude reaches the peak current of 500A, the measured result is about 5.2% smaller than the FEA result. The error rate of measured output torque increases with increase of driving current amplitude. This is mainly due to the fact that during finite element simulation, the temperature of permanent magnet was set to 90℃, but during experimental testing, the temperature of permanent magnet was not constant. The larger the driving current, the greater increase in copper and iron losses, and the higher temperature rise of permanent magnet and stator winding. However, the temperature rise leads to a decrease in residual magnetization and air gap flux density of permanent magnet. Compared with the finite element simulation results, the decrease of measured output torque increases with increase of driving current.

The FEA and actual measurement results of steady torque of the prototype under rated operating conditions is shown in Fig. 31. From Fig. 31, it can be seen that the steady torque trend of FEA and actual testing under rated operating conditions is basically consistent, but there is a certain phase difference. Compared with the result of FEA, the steady torque measured under rated conditions decreased by about 3%.

Prototype external characteristic testing

The external characteristic test of prototype is to test the output torque and power of prototype at different speeds under flux weakening control strategy. When testing the external characteristics of the prototype, the prototype operates in an electric state with a rated driving current amplitude of 260A. The prototype speed increases from 3000 to 13000 rpm with a step size of 1000 rpm. At different speeds, the corresponding load of the prototype is provided by a dynamometer, with a load step interval of 15Nm, to test the output torque and power at different speeds. The comparison of external characteristics between FEA and actual measurement of the prototype is shown in Fig. 32.

From Fig. 32, it can be seen that when the speed is between 3000 and 7000 rpm, the prototype maintains a constant torque output, and the output power increases linearly with increase of speed; When the speed is 7000 ~ 10000 rpm, the output torque of prototype decreases with increase of speed, and basically maintains a constant power output. When the speed is 10,000 ~ 13000 rpm, the output power of the prototype decreases with increase of speed. The measured output torque and output power at different speeds are both lower than FEA values. The higher the speed, the greater the error, because the higher speed, the greater iron loss and eddy loss of the permanent magnet, resulting in a higher temperature rise of the permanent magnet. The greater the decrease



(a) Rotor assembly diagram



(b) stator winding

Fig. 28. Rotor and stator of the prototype.

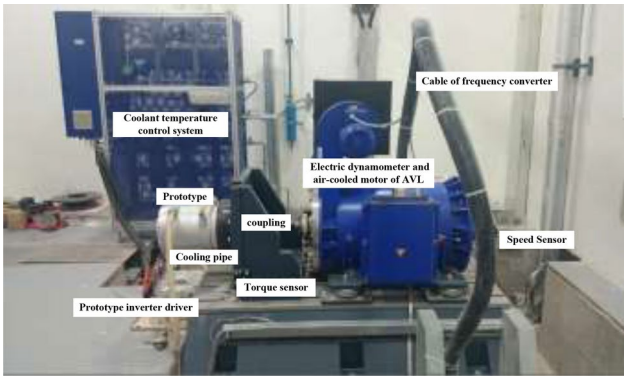


Fig. 29. Experimental testing bench.

in residual flux density and air gap flux density of the permanent magnet, the greater the reduction in output torque and power. At a prototype speed of 13000 rpm, the maximum error between the measured output torque and power and the result of FEA is 5%. When the speed is 9000 rpm, the measured maximum output power is 132 kW. The measured values of output torque and power shown in Fig. 26 are basically consistent with the simulation results of FEA, verifying the accuracy of FEA calculations during the design of IPMSM.

Conclusion

An IPMSM with I2V type rotor topology is proposed, and the torque performance including output torque ripple, average torque, and torque density, etc. have been analyzed and compared with typical V and IV types. The results showed that IPMSM with I2V type has better torque performance than the other two with slightly

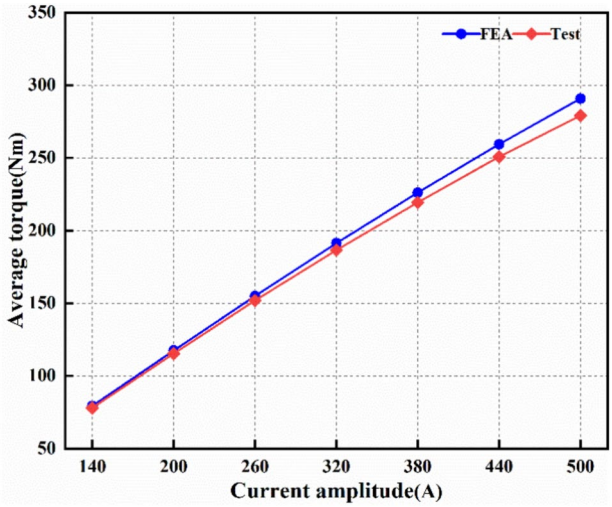


Fig. 30. Output torque comparison between FEA and testing.

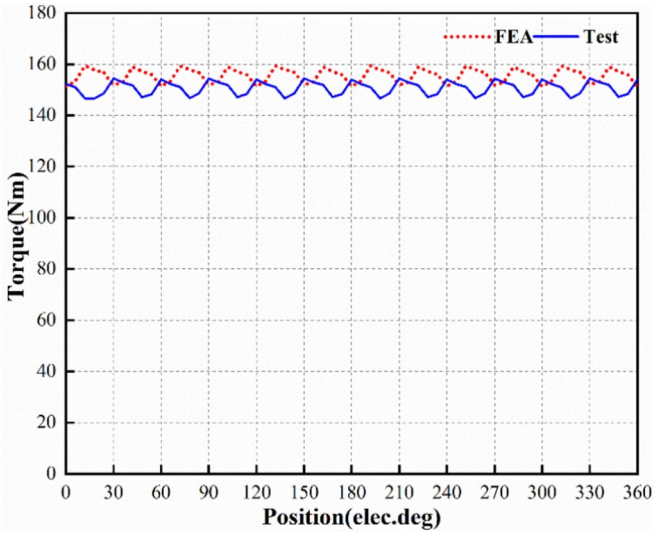


Fig. 31. Steady torque comparison between FEA and testing.

reduced permanent magnet usage. By adding a set of circular magnetic isolation holes between the second and third layers of permanent magnets in each pole of the rotor, a parameterized model of circular magnetic isolation holes is established to optimize and adjust the position and diameter of the set of circular magnetic isolation holes to further optimize the magnetic circuit, change the local magnetic flux density distribution in the air gap, and significantly reduce the torque ripple of the IPMSM with I2V type. The optimized I2V type has a torque ripple reduction ratio of 67.3% under rated operating conditions. We processed and produced an I2V prototype and built a prototype experimental platform to measure the torque and external characteristics of the IPMSM. The measured results are basically consistent with FEA results, verifying the effectiveness of analysis method and simulation results in this paper.

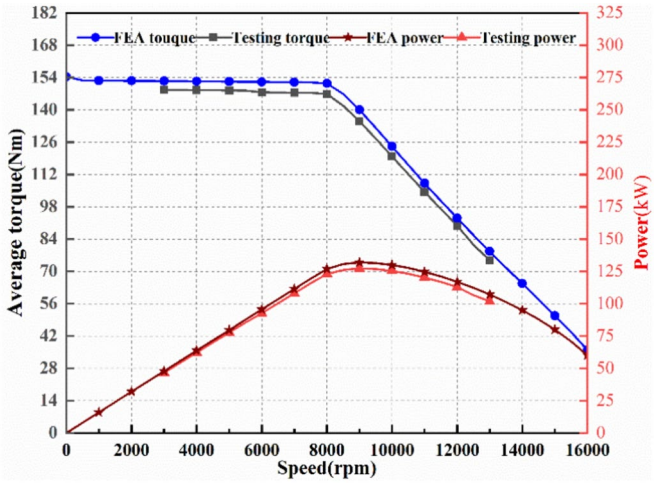


Fig. 32. External characteristics comparison between FEA and testing.

Data availability

All data generated or analysed during this study are included in this published article.

Received: 18 January 2025; Accepted: 5 March 2025
Published online: 14 March 2025

References

1. Pellegrino, G. et al. Comparison of induction and PM synchronous motor drives for EV application including design examples. *IEEE Trans. Ind. Appl.* **48**(6), 2322–2332 (2012).
2. Sun, X. et al. Core losses analysis of a novel 16/10 segmented rotor switched reluctance BSG motor for HEVs using nonlinear lumped parameter equivalent circuit model. *IEEE/ASME Trans. Mechatron.* **23**(2), 747–757 (2018).
3. Sun, X. et al. Study on segmented-rotor switched reluctance motors with different rotor pole numbers for BSG system of hybrid electric vehicles. *IEEE Trans. Veh. Technol.* **68**(6), 5537–5547 (2019).
4. Sun, X. et al. MPTC for PMSMs of EVs with multi-motor driven system considering optimal energy allocation. *IEEE Trans. Magn.* **55**(7), 1–6 (2019).
5. Yang, Y. et al. Design and comparison of interior permanent magnet motor topologies for traction applications. *IEEE Trans. Transp. Electr.* **3**(1), 86–97 (2016).
6. Momen, F., Rahman, K. & Son, Y. Electrical propulsion system design of Chevrolet Bolt battery electric vehicle. *IEEE Trans. Ind. Appl.* **55**(1), 376–384 (2018).
7. Shen, Y. & Zhu, Z. Q. Analytical prediction of optimal split ratio for fractional-slot external rotor PM brushless machines. *IEEE Trans. Magn.* **47**(10), 4187–4190 (2011).
8. Han, S. H., Jahns, T. M. & Zhu, Z. Q. Design tradeoffs between stator core loss and torque ripple in IPM machines. *IEEE Trans. Ind. Appl.* **46**(1), 187–195 (2010).
9. Chen, X. et al. A high-fidelity and computationally efficient model for interior permanent-magnet machines considering the magnetic saturation, spatial harmonics, and iron loss effect. *IEEE Trans. Industr. Electron.* **62**(7), 4044–4055 (2015).
10. Hu, W. Y. *Performance Optimization of Interior Permanent Magnet Synchronous Motor with Different Rotor Topologies* (Beijing Jiao tong University, 2019).
11. Hu, Y. H. *Research on Interior Permanent Magnet Synchronous Machines for EV Applications* (Nanjing University of Aeronautics and Astronautics, 2017).
12. Hu, Q. G. & Wu, M. Q. Effect of segmented shift of rotor magnetic pole on torque of permanent magnet synchronous motor. *J. Huaqiao Univ. (Nat. Sci.)* **43**(01), 29–35 (2022).
13. Shi, Z. et al. Torque analysis and dynamic performance improvement of a PMSM for EVs by skew angle optimization. *IEEE Trans. Appl. Supercond.* **29**(2), 1–5 (2018).
14. Ren, W. et al. Reduction of cogging torque and torque ripple in interior PM machines with asymmetrical V-type rotor design. *IEEE Trans. Magn.* **52**(7), 1–5 (2016).
15. Lu, X. et al. Investigation of permanent-magnet motor drives incorporating damper bars for electrified vehicles. *IEEE Trans. Industr. Electron.* **62**(5), 3234–3244 (2014).
16. Barman, D. & Pillay, P. Effect of skewing in a variable flux interior permanent magnet synchronous machine. *IEEE Trans. Ind. Appl.* **56**(6), 6399–6410 (2020).
17. Wang, S. et al. Filling force valley with interpoles for pole-frequency vibration reduction in surface-mounted PM synchronous machines. *IEEE Trans. Industr. Electron.* **67**(8), 6709–6720 (2019).
18. Peng, C. et al. Torque ripple and electromagnetic vibration suppression in permanent magnet synchronous motor using segmented rotor with different pole widths. *IEEE Trans. Magn.* **58**(9), 1–5 (2022).
19. Wang, G., Li, N. & Zhao, D. Strength analysis of motor flange structure based on finite element method. *Electr. Mach. Control Appl.* **2**, 72–75 (2020).
20. Fang, H. et al. Rotor design for high speed high power permanent magnet synchronous machines. *IEEE Trans. Ind. Appl.* **53**(4), 3411–3419 (2017).
21. Zhang, F. G. et al. Rotor retaining sleeve design for a 1.12 MW high-speed PM machines. *IEEE Trans. Ind. Appl.* **51**(5), 3675–3685 (2015).

Author contributions

Lianbo Niu wrote the main manuscript text, Lianbo Niu prepared figures， All authors reviewed the manuscript.

Competing interests
The authors declare no competing interests.

Additional information
Correspondence and requests for materials should be addressed to L.N.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

© The Author(s) 2025